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Assignment - 2 (Astroparticle physics)

① we observe non counts in a source region. we assume these counts follow a Poisson distribution with an expected value μ composed of a ~~sig~~ signal component s and a known background component n_{bkg}.

$$\mu = s + n_{bkg}$$

The likelihood fn $L(s)$ for observing non counts given the parameters is,

$$L(s) = P(n_{on} | s, n_{bkg})$$

$$= \frac{(s+n_{bkg})^{n_{on}} e^{-(s+n_{bkg})}}{n_{on}!}$$

To find the MLE for s, we take the natural log to obtain the log-likelihood $\ell(s)$

$$\ell(s) = \ln L(s) = n_{on} \ln(s+n_{bkg}) - (s+n_{bkg}) - \ln(n_{on}!)$$

$$\frac{d\ell}{ds} = \frac{n_{on}}{s+n_{bkg}} - 1 = 0$$

$$s + n_{bkg} = n_{on}$$

$$\boxed{\hat{s} = n_{on} - n_{bkg}}$$

ⓐ $n_{on} = 10$, $n_{bkg} = 3$.

excess signal $\hat{s} = 10 - 3 = 7$

Significance of detection:

We use the likelihood ratio test. The test statistic TS is given by,

$$TS = 2[\,\ell(\hat{s}) - \ell(0)\,] = 2\left[\, n_{on} \ln\left(\frac{n_{on}}{n_{bkg}}\right) - (n_{on} - n_{bkg}) \right]$$

$$= 2\left[\, 10 \ln\left(\frac{10}{3}\right) - 7 \right]$$

$$= 10.08$$

The significance in standard deviations is

$$Z = \sqrt{TS} \simeq 3.176.$$

Confidence intervals:

The $k\sigma$ confidence interval is found by solving

$$\ell(\hat{s}) - \ell(s) = \frac{1}{2} k^2.$$

1σ ($k=1$, $\Delta\ell = 0.5$) $s \in [4.16, 10.50]$

2σ ($k=2$, $\Delta\ell = 2$) $s \in [1.93, 14.72]$

$n_{on} = 5$; $n_{bkg} = 3$

excess signal, $\hat{s} = 5 - 3 = 2$

$$TS = 2\left[\, 5 \ln(5/3) - (5-3) \right] \simeq 1.11$$

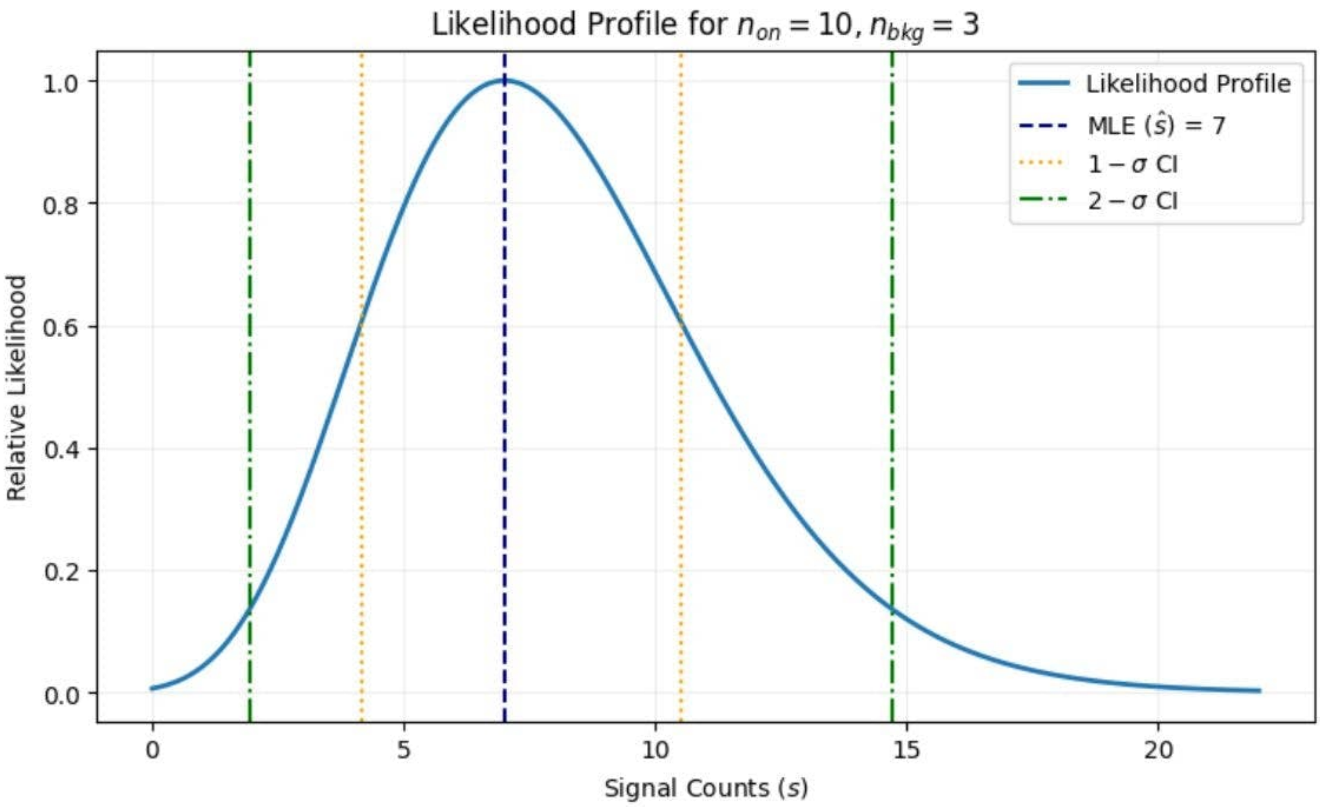
$Z = \sqrt{1.11} = 1.056$. This indicates the signal is not statistically significant.

5σ upper limit.

For an upper limit at 5σ ($k=5$), we solve for $s > \hat{s}$ such that $\ell(\hat{s}) - \ell(s) = \frac{1}{2}(5^2) = 12.5$.

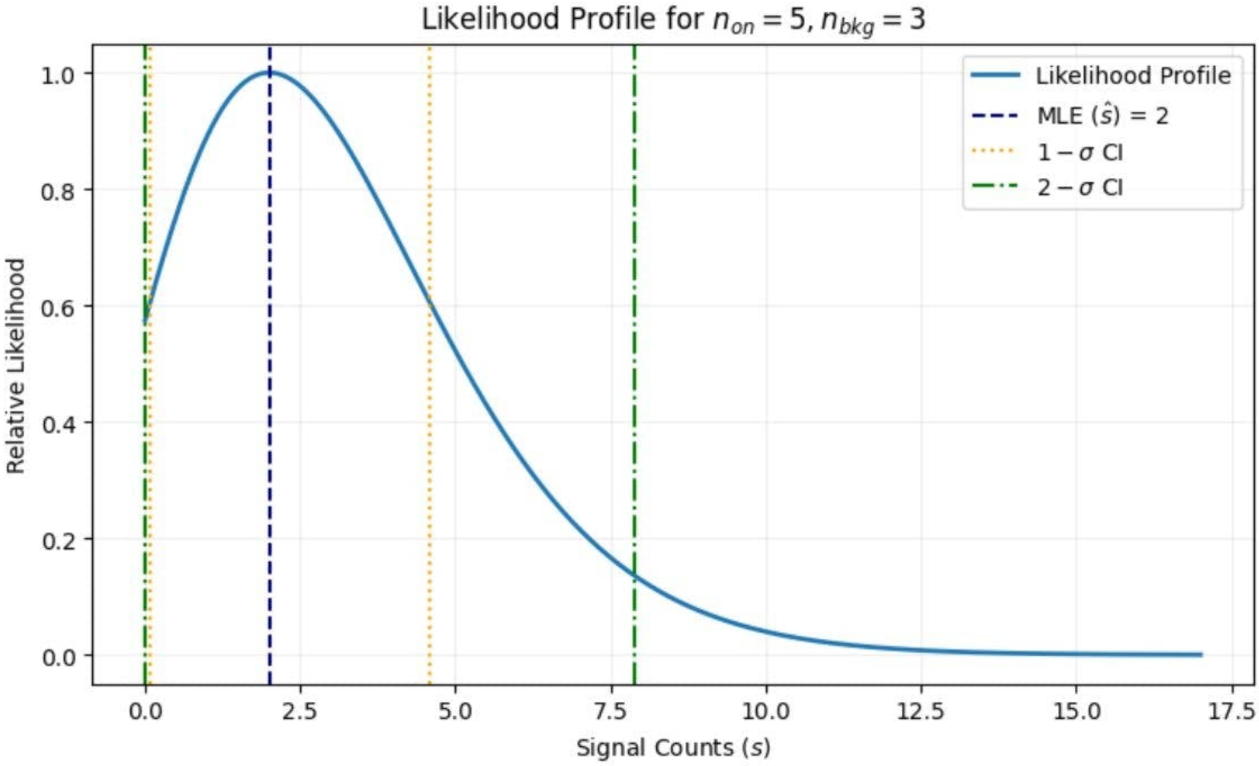
$$5 \ln(s+3) - (s+3) = 5 \ln 5 - 5 - 12.5$$

Solving numerically, the 5σ upper limit $\simeq 22.68$



Likelihood Profile for $n_{on} = 10, n_{bkg} = 3$
Likelihood Profile
MLE ($\hat{s}$) = 7
$1-\sigma$ CI
$2-\sigma$ CI
Relative Likelihood
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Signal Counts (s)

Signal Counts (s)



(2) In many astroparticle physics experiments, the exact level of the background noise during a source observation (t_{on}) is not known before hand. To account for this, we perform a separate measurement in a background control region (t_{off}) where no source is expected to be present.

Let N_{on} and N_{off} represent the observed counts in the respective regions. we define s as the expected signal counts and b_{off} as the expected background in the off region. Given as acceptance ratio, $\alpha = \frac{a_{on}}{a_{off}}$, the expected background in the on-region is $b_{on} = \alpha b_{off}$

The joint prob. of observing our data follows a product of two poisson distributions:

$$L(s, b_{off}) = \frac{(s + \alpha b_{off})^{N_{on}} e^{-(s+\alpha b_{off})}}{N_{on}!} \times \frac{b_{off}^{N_{off}} e^{-b_{off}}}{N_{off}!}$$

to simplify minimization we define the negative log-likelihood ($\ell = -\ln L$).

$$\ell(s, b_{off}) = (s + \alpha b_{off}) - N_{on} \ln(s + \alpha b_{off})$$
$$+ b_{off} - N_{off} \ln b_{off} + c$$

for signal $\frac{\partial \ell}{\partial s} = 1 - \frac{N_{on}}{\hat{s} + \alpha b_{off}} = 0$

$$\hat{s} + \alpha b_{off} = N_{on}$$

for background, b_{off}: $\frac{\partial \ell}{\partial b_{off}} = \alpha - \frac{\alpha N_{on}}{\hat{s} + \alpha b_{off}} + 1 - \frac{N_{off}}{N_{off}} = 0$

substituting 1st eq in 2nd,

$$\hat{b}_{off} = N_{off}.$$

so, $$\boxed{\hat{s} = N_{on} - \alpha N_{off}}$$

The significance of a detection is then determined by the test statistic μ, which compares the global maximum likelihood to the likelihood under the background-only hypothesis ($s=0$).

$$\mu = 2\left[l(s=0, \hat{b}_{off,0}) - l(\hat{s}, \hat{b}_{off}) \right]$$

- The primary distinction between the c stat and w stat lies in the treatment of background uncertainty.

In c stat the background is treated as a known constant provided by a model, meaning the only source of error is the poisson fluctuation of the signal region itself.

In w stat, we acknowledge that our background estimate is itself a measurement with its own poisson error. By including b as a nuisance parameter and profiling over it we naturaly incorporate the uncertainty of the background measurement into our final results.

Consequently, w stat typicaly produces lower detection significances and broader confidence intervals compared to c stat, as it correcty accounts for the fact that a high count in the 'on' region could partly be explained by a random upward fluctuation of the background.

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--- Case 1: n_on=10, n_off=6 ---
Excess (s): 7.00
Significance: 2.38 sigma
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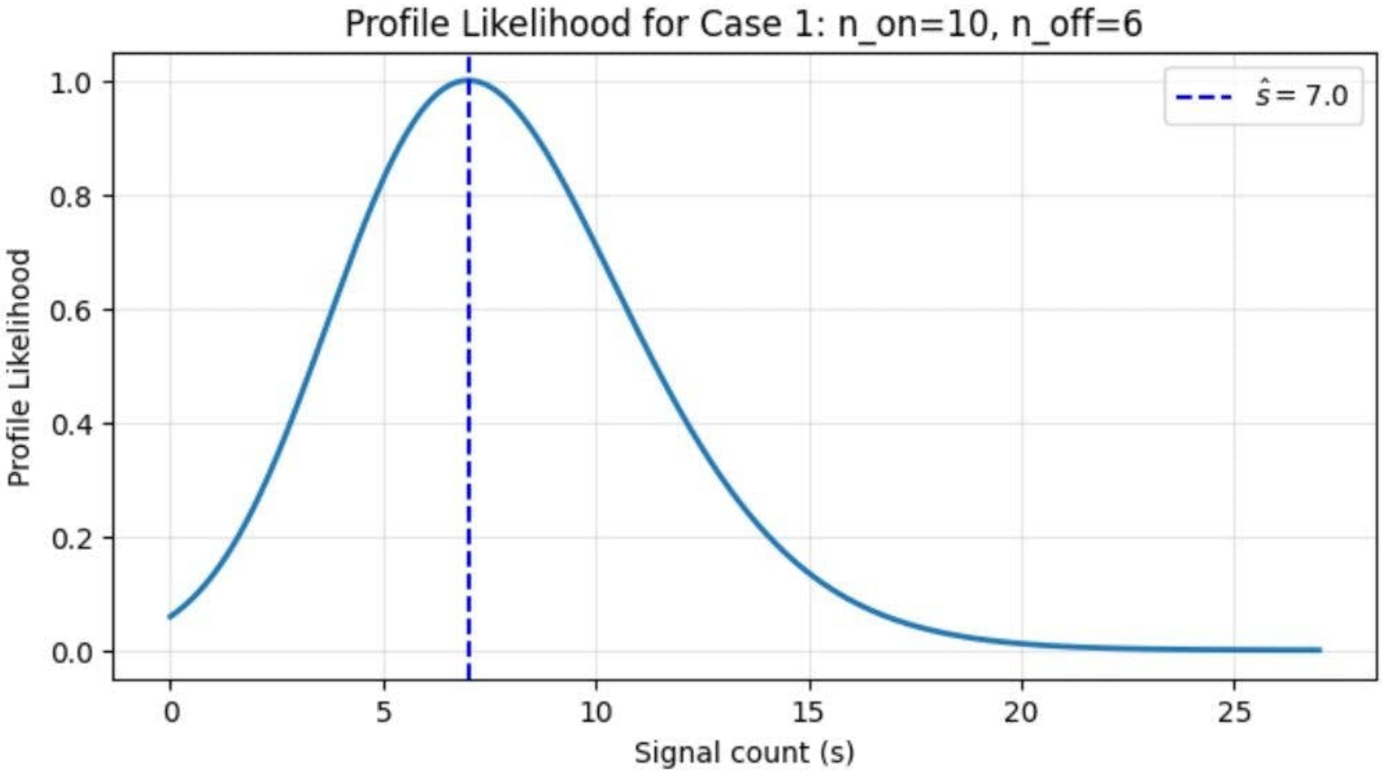

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--- Case 2: n_on=5, n_off=6 ---
Excess (s): 2.00
Significance: 0.83 sigma
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